

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Critical Line List-AKABEL
Document Number	EA-BFPCL-GPMC-GGPL-FEED-MPP-LST-013
Document Revision	R01
Document Status	Issued For Review
Originator	Obinna Joseph
Security Classification	Restricted
Issue Date	2 Aug 2022

Critical Line List-AKABEL

R01	02.08.22	Issued For Review	O. Joseph	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	02.08.22	Issued for Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-LST-013

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

Critical Line List-AKABEL

Document No.	Description

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Critical Line List-AKABEL

GENERAL NOTES

1. In accordance with DEP 31.38.01.26-Gen, the following in-plant process lines shall be considered for a comprehensive piping stress analysis using Industry recognised Stress analysis software. Bentley AutoPIPE CONNECT v11.2 will be used.
 - Process lines size DN 80 and above (NPS 3 and above) subjected to slug flow or water hammer
 - Process lines size of size DN 80 and above (NPS 3 and above) subjected to two phase flow
 - Process lines size DN 50 (NPS 2) and above subjected to relief loads;
 - Process lines size DN 300 (NPS 12) with design temperatures from 40 °C (100 °F) to 175 °C (350 °F)
 - Blowdown and flare header or vent piping, but not atmospheric vent piping
 - Lines having substantial concentrated loads such as valves, fittings, unsupported vertical risers and branches
 - Pipe size DN 500 (NPS 20) and larger
2. The formal computerised analysis shall provide report in chart methods that provide an evaluation of the forces, moments and stresses caused by displacement strains.
3. The integrity and flexibility of lines that are not on the critical line list ("non-critical") shall be guaranteed and shall be properly supported.
4. All critical lines shall be identified as such on the project line-list.
6. Stress analysis of the main product pipeline section, from pig receiver to pig launcher shall be conducted separately by pipeline discipline When full survey information/Data on the Right of Way (ROW) is available.

Critical Line List-AKABEL

S/No.	LINE NUMBER DESCRIPTION						LINE NUMBER	MATERIAL	OPERATING COND.		DESIGN CONDITION			SERVICE	LINE ORIGIN		LINE DESTINATION		REMARKS
	System	LOCATION CODE	NOMINAL LINE SIZE	LINE SERVICE DESIGNATION	SERIAL NUMBER	PIPING CLASS			TEMP. (°C)	PRES. (bar.g)	DES. TEMP (°C)	DES. PRES. (bar.g)	HYDROTEST PRESS. (bar.g)		ITEM OR LINE NO. FROM	P&ID DRAWING NO	ITEM OR LINE NO. TO	P&ID DRAWING NO	
1	Process System	AKABP	36"	P	012	CL900	AKABP 36" P 012 CL900	ASME B36.10, API SPEC 5L GRADE X70 PSL 2	70	100	-29 / 100	110	165	Multiphase(Gas+condensate)	SDV 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	AKAB-A-100 (PIG RECEIVER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	
2		AKABP	36"	P	013	C06	AKABP 36" P 013 C06	ASME B36.10, API SPEC 5L GRADE X60 PSL 2	70	100	-29 / 100	110	165	Multiphase(Gas+condensate)	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	TP 102 - SLUG CATCHER AT AKABEL	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	
3		AKABP	4"	P	015	C06	AKABP 4" P 015 C06	ASME B36.10, API SPEC 5L GRADE B PSL 2	70	100	-29 / 100	110	165	Multiphase(Gas+condensate)	AKAB-A-100 (PIG RECEIVER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	AKABP-12"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	
4		AKABP	12"	P	014	C06	AKABP 12" P 014 C06	ASME B36.10, API SPEC 5L GRADE B PSL 2	70	100	-29 / 100	110	165	Multiphase(Gas+condensate)	AKAB-A-100 (PIG RECEIVER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	AKABP-36"-P-013-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	
5		AKABP	4"	P	016	C06	AKABP 4" P 016 C06	ASME B36.10, API SPEC 5L GRADE B PSL 2	70	100	-29 / 100	110	165	Multiphase(Gas+condensate)	AKAB-A-100 (PIG RECEIVER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	PSV 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	
6	Pressure Relief & Blowdown System	AKABP	6"	B	001	B06	AKABP 6" B 001 B06	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	AKABP-36"-P-012-X70-CL900	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	TP 101 - VENT SYSTEM AT AKABEL	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	
7		AKABP	6"	B	005	B06	AKABP 6" B 005 B06	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	PSV 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	AKABP-6"-B-001-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	
8	Closed & Open Drain System	AKABP	4"	D	002	C06	AKABP 4" D 002 C06	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	100	-29 / 100	110	165	Drain	AKAB-A-100 (PIG RECEIVER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	TP 103 - COLLECTION DRAIN SYSTEM AT AKABEL	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_08	